

HR Shortlisting Agent

AI-Powered Recruitment Evaluation System

INTERNSHIP PROJECT • TASK 1

The Problem

- ✦ HR teams spend hours manually screening resumes.
- ✦ High risk of unconscious bias and fatigue.
- ✦ Inconsistent scoring across different recruiters.

The Solution

- ✦ Automated evaluation using High-Performance LLMs.
- ✦ Structured scoring rubric based on 5 key dimensions.
- ✦ **Human-in-the-Loop** : HR retains final control.

Technical Stack

- ✦ **UI** : Streamlit (Premium Glassmorphism UI)
- ✦ **Engine** : LangChain & Pydantic
- ✦ **LLM** : Llama 3.1 70B (via Groq LPU™)
- ✦ **Security** : python-dotenv & Pydantic Validation

The Scoring Rubric

- ✦ **Skills Match (30%)** : Technical competency.
- ✦ **Experience (25%)** : Relevance and seniority.
- ✦ **Projects (20%)** : Quality of portfolio.
- ✦ **Education (15%)** : Qualification match.
- ✦ **Communication (10%)** : Resume clarity.

Human-in-the-Loop

- ✦ Manual score correction interface.
- ✦ Automatic recalculation of ranking.
- ✦ Detailed audit logs for transparency.

Security & Reliability

- ✦ **Prompt Injection Protection** : Schema-enforced.
- ✦ **PII Privacy** : No persistent personal storage.
- ✦ **Determinism** : Temperature=0 for consistency.

Key Architecture

- ✦ **Ingest** : PDF/TXT parsing via pypdf.
- ✦ **Extract** : Structured requirement extraction.
- ✦ **Score** : Individual rubric dimension grading.
- ✦ **Report** : HTML/JSON export generation.

Demo Results

- ✦ Successfully processed 5 diverse resumes.
- ✦ 100% accuracy in identifying skill gaps.
- ✦ Near-instant results (<1 second per resume).

Future Roadmap

- ✦ Integration with LinkedIn Scraping API.
- ✦ Automated interview scheduling (Task 2).
- ✦ Direct integration with corporate ATS systems.